METRIC
A-A-52444A
February 15, 2011
SUPERSEDING
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August 6, 1993

COMMERCIAL ITEM DESCRIPTION

GASKETS: COMBINATION OF METAL AND NONASBESTOS FACING MATERIAL (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-G-14243D, which is canceled.

1. SCOPE. This CID covers three types of combination gaskets made from a combination of metal and nonasbestos facing material. These combination gaskets are used in automotive vehicle engine applications where they will be subjected to high temperatures as in cylinder heads and exhaust manifolds. The combination gaskets are referred to herein as "gaskets".

2. CLASSIFICATION. The gaskets shall be of the following types (see 7.2):

Type I - Steel sheet with nonasbestos on both surfaces.
Type II - Steel sheet with nonasbestos on one surface.
Type III - Nonasbestos sheathed in sheet metal.

3. SALIENT CHARACTERISTICS

3.1 Materials. Unless otherwise specified herein, the materials used to make the gaskets shall be in accordance with the manufacturer's materials specifications. The use of recovered material made in compliance to regulatory requirements is acceptable provided all the requirements of this CID are met (see 4).

3.1.1 Steel sheet. Steel sheet shall conform to ASTM A1008/ A1008M for low carbon (0.15 percent (%) maximum), cold-rolled, commercial quality steel.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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3.1.2 Steel strip. Steel strip shall conform to ASTM A109/ A109M for low carbon (0.15% maximum), cold-rolled, temper 4 or 5, matte (dull) finish steel.

3.1.3 Long terne steel sheet. Long terne steel sheet shall conform to ASTM A308/ A308M for commercial quality, oiled, coating designation LT35 steel.

3.1.4 Tin plate. Tin plate shall conform to ASTM A623 for steel as base metal (Type-optional), tin coating weight designation 25, temper designation T-4, box annealed tin plate.

3.1.5 Copper sheet. Copper sheet shall conform to ASTM B152/ B152M for cold-rolled, soft anneal designation O60 copper.

3.1.6 Other metals. When specified (see 7.2), other metals may be used in the fabrication of gaskets.

3.2 Design and construction. The details of design and construction (e.g., configuration, size, dimensions) shall be in accordance with the applicable engineering drawing (AED) (see 7.2).

3.2.1 Type I gaskets. Type I gaskets shall be made from steel, sheet or strip, as specified under 3.1.1 through 3.1.4 and nonasbestos material on both surfaces. Type I gaskets shall have the steel sheet recessed and formed so as to have closely spaced tangs projecting from both surfaces. The tangs shall hold the nonasbestos materials to the metal after being pressed. When required to prevent delamination, adhesives may be used. The nonasbestos material shall be plied evenly on both surfaces of the metal with the same thickness on each surface.

3.2.2 Type II gaskets. Type II gaskets shall be made from steel sheet as specified in 3.1.3 or 3.1.4 and nonasbestos material on one surface. The type II gaskets shall have the steel sheet recessed and formed so as to have closely spaced tangs projecting from one surface only. The tangs shall hold the nonasbestos material to the metal after being pressed. When required to prevent delamination, adhesives may be used. The nonasbestos material shall be plied evenly on one surface and shall be uniform in thickness. The sheet metal shall be formed to protect the outer edges of the gasket but shall not be closed over the nonasbestos material at the outer edges.

3.2.3 Type III gaskets. Type III gaskets shall be made from nonasbestos material sheathed on both faces with metallic sheets as specified in 3.1.3, 3.1.4 or 3.1.5. Metallic grommets shall band the gasket, as specified in the AED. Adhesives shall not be used. The metallic sheets shall be formed to protect the outer edges of the gasket but shall not be closed over the nonasbestos material at the outer edges.

3.2.4 Grommets. Grommets used to fabricate type III gaskets shall be made of material as specified in the AED and, if not specified, in accordance with the manufacturer's specifications/ drawings.

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3.2.5 Coating. To prevent seizure to metallic surfaces, a coating of pulverized carbon containing a binder, or equivalent material, shall be evenly applied to the nonasbestos material surfaces of type I and type II gaskets. Unless otherwise specified in the AED, type III gaskets shall be coated with lacquer or varnish.

3.3 Performance.

3.3.1 Compressibility. The gaskets shall be capable of withstanding a 13.8 megapascals (MPa) load without rupture and compress 10 to 20% from original thickness.

3.3.2 Liquid absorption. The nonasbestos gasket material shall show no increase in weight in excess of 20% after being immersed for 22 hours and shall show no evidence of disintegration or of becoming soft or spongy after immersion. This shall be verified using the immersion fluids and test procedures of ASTM F146.

3.3.3 Resistance to boiling water. When immersed in boiling water for 2 hours, the nonasbestos material shall not become soft or spongy and shall not separate from the metal sheet.

3.3.4 Moisture loss. The moisture loss of the nonasbestos material shall not be more than two percent by weight when subjected to a temperature of 82 degrees Celsius (°C) until weight becomes constant.

3.3.5 Flexibility. The metal and nonasbestos material in type I and type II gaskets shall not separate when bent around a 50.8 mm mandrel.

3.4 Identification and marking. Unless otherwise specified in the AED, identification and marking of gaskets shall be permanent and legible and shall include, as a minimum, the AED part number and the manufacturer's CAGE code and part number (see 7.2).

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Responsibility for inspection. The contractor is responsible for all inspections, including examinations and tests.

5.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the

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first delivery and thereafter as may be otherwise provided for under the provisions of the contract or order.

5.3 Product Conformance . The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.2).

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Addresses for obtaining copies of non-Government publications. Copies of ASTM A109/ A109M "Standard Specification for Steel, Strip, Carbon (0.25 Maximum Percent), Cold-Rolled", ASTM A308/ A308M "Standard Specification for Steel Sheet, Terne (Lead-Tin Alloy) Coated by the Hot-Dip Process", ASTM A623 "Standard Specification for Tin Mill Products, General Requirements", ASTM A1008/ A1008M "Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable", ASTM B152/ B152M "Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar" and ASTM F146 "Standard Test Methods for Fluid Resistance of Gasket Materials" are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. If required, the specific issue of individual documents referenced.
c. Type of gasket required (see 2).
d. Title, number, and date of AED and part number (see 3.2 and 3.4).
e. Metal required if other than as specified (see 3.1.6).
f. Selection of applicable level and packaging requirements (see 6).

7.3 Cross-reference data. Gaskets conforming to this CID are interchangeable with gaskets conforming to MIL-G-14243D, dated 26 April 1989.

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